

Meeting Minutes 3

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)

Meeting Minutes 3A (In-Class – With Professor)

(Dataset Planning & Socioeconomic Context Discussion)

Date and Time

20 Jan 2026, 9:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Review dataset expansion plan
- Discuss socioeconomic context
- Align datasets with research problem

Discussion Points

- Manjunath explained the plan to enhance CMS Hospital Cost Report data using public datasets capturing labor costs and geographic context.
- The team informed the professor that **U.S. Census data** would be used to provide **socioeconomic context**, particularly income-related variables.
- Professor Maheshwari agreed that socioeconomic context is relevant but emphasized that **all datasets must directly support the research question or problem statement**.
- She cautioned against adding variables based solely on availability and stressed the need for clear justification in the methodology.

Decisions Made

- Census data approved in principle for socioeconomic context.
- Proceed with non-Census datasets for baseline ML models.
- Maintain a phased dataset integration approach.

Action Items

- Implement approved datasets (BLS wage data and Urban–Rural codes).
- Deliver baseline ML models before expanding features.

Any concerns (if any)

- Risk of scope expansion without clear research justification.

Next Meeting Date and Time

22 Jan 2026 (Group meeting – implementation)

Minutes Prepared By

Manjunath Muthineni

Meeting Minutes 3B (Group Meeting – Separate)

(Implementation of BLS Wage Data & Urban–Rural Continuum Codes)

Date and Time

22 Jan 2026, 7:00 PM – 9:00 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Implement BLS healthcare wage data
- Implement Urban–Rural Continuum Codes
- Validate feature engineering and joins

Discussion Points

- The group confirmed that **only hospital-based healthcare occupation codes** were used from the BLS data to align wages with hospital cost structures.
- Abhishekh explained how BLS data across multiple years was standardized and how **employment-weighted average healthcare wages** were calculated at the state-year level.
- Urban–Rural Continuum Codes (RUCC) were processed and simplified into a **binary rural vs urban classification** for modeling clarity.
- CMS Hospital Cost Report data was confirmed as the anchor dataset for all joins.
- Validation checks were discussed to ensure no duplication, leakage, or key mismatches.

Decisions Made

- Hospital-based healthcare wages approved as the labor cost proxy.
- Employment-weighted wage selected as the final wage feature.
- RUCC simplified into rural vs urban classification.

Action Items

- Finalize CMS–BLS–RUCC merged dataset.
- Prepare dataset for baseline ML modeling.

Any concerns (if any)

- Risk of misalignment if non-hospital occupations are included; mitigated through strict filtering.

Next Meeting Date and Time

27 Jan 2026 (In-class meeting with professor)

Minutes Prepared By

Abhishekh Choudhary

Meeting Minutes 4

Meeting Minutes 4A (In-Class – With Professor)

(Baseline ML Model Review & Census Data Access Guidance)

Date and Time

27 Jan 2026, 9:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)

Absent

- Krishna Chaitanya Venuturumilli (0874277)

Agenda

- Present baseline ML models
- Review model performance and evaluation
- Discuss data gaps and next steps

Discussion Points

- Manjunath presented baseline ML models predicting **Cost-to-Charge Ratio (CCR)** and **Profit Margin** using CMS, BLS, and rural–urban features.
- The modeling pipeline and evaluation approach were reviewed, focusing on **interpretability and methodological correctness** rather than absolute accuracy.
- The team confirmed that **Census data had not yet been integrated**, as priority was given to validating baseline ML models before expanding the feature set.
- The group explained difficulties in locating and extracting Census data in a reproducible format.
- Professor Maheshwari clarified that Census data visible in tools such as Tableau is **not suitable for academic extraction**.
- She advised using **official U.S. Census APIs** to obtain income-related variables if they strengthen the research explanation.
- The professor confirmed that the **baseline models were methodologically sound** as a first iteration.

Decisions Made

- Baseline ML models approved as Iteration 1.
- Census data approved for Iteration 2 using API-based access.

Action Items

- Explore Census API for median household income.
- Retrain models after Census integration and compare results.

Any concerns (if any)

- Ensuring Census data improves explanatory value rather than adding noise.

Next Meeting Date and Time

29 Jan 2026 (Group meeting – Census integration)

Minutes Prepared By

Rajesh Thota

Meeting Minutes 4B (Group Meeting – Separate)

(Census Integration & Dataset Finalization)

Date and Time

29 Jan 2026, 7:00 PM – 8:30 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Integrate Census median household income
- Validate joins and assumptions
- Finalize dataset

Discussion Points

- Census median household income was integrated using **API-sourced data** to provide socioeconomic context.
- Abhishekh validated joins to ensure no duplication or leakage.
- Neha confirmed income as a socioeconomic proxy rather than a direct hospital performance variable.
- The group agreed to **freeze the dataset** after this integration.

Decisions Made

- Census median household income approved as final feature.
- Dataset expansion closed.

Action Items

- Retrain final models.
- Update documentation and results.

Any concerns (if any)

- None

Next Meeting Date and Time

3 Feb 2026 (In-class meeting with professor)

Minutes Prepared By

Krishna Chaitanya Venuturumilli